Quantum preparations select continuations at a bounded Newton cusp

Result, 2026-10-27; written derivation, unrefereed. For the bounded C^1 cusp potential already used in the preparation-floor note, there are two explicit coherent-state families with the **same position density** at $t = 0$, the same limiting phase-space point $(0, 0)$, and the same limiting incoming trajectory for $t < 0$. Their unique Schrödinger evolutions have different outgoing semiclassical limits for $t > 0$: one rebounds on the positive side and the other crosses to the negative side. The preparations differ by a momentum shift tending to zero. Their normalized superpositions yield any prescribed mixture of the two outgoing point measures.

The result is local in time and uses the specified logarithmically shrinking preparation. It neither classifies every possible preparation nor selects a waiting-time distribution. It shows explicitly that the limiting incoming phase-space history alone does not determine the outgoing limit of these quantum families. The positive value of $\hbar$ is supplied, not derived.

This phenomenon has substantial prior art. Athanassoulis–Paul, Theorem 2.1 and Lemma 5.3 (theorem statements and relevant passages read) treat splitting and one-dimensional cutoff $-|q|^{1+\theta}$ potentials, including $\theta = 1/2$, with detailed assumptions on phase-space preparations. Chabu, Theorem 1.12 and Proposition 3.1 (statements and surrounding discussion read) exhibit the same incoming semiclassical history with distinct outgoing histories, and mixtures, for the conical potential $-|q|$. In Theorem 1.12 the two initial profiles can be chosen equal, so identical initial position densities also occur in that conical example. The present calculation uses the repository's smoother C^1 cusp and explicit Gaussian preparations; exact coverage by the cited rough-potential theorems is not settled and no literature priority is claimed.

1. Potential, initial data and target trajectories

Use dimensionless calibrated units, $m = 1$, and choose $R > 0$ and a smooth even cutoff χ equal to one on $[-R, R]$ and zero outside $[-2R, 2R]$. Put

$$V(q) = -\frac{2}{3}|q|^{3/2}\chi(q), \qquad F(q) = -V'(q), \qquad H_\hbar = -\frac{\hbar^2}{2}\partial_q^2 + V(Q). \tag{1}$$

V and F are bounded, $V \in C^1$, and on $|q| \le R$, $F(q) = \operatorname{sgn}(q)\sqrt{|q|}$. The bounded-perturbation theorem gives a self-adjoint $H_\hbar$ on $H^2(\mathbb{R})$ for every $\hbar > 0$. Write $F_* = \|F\|_\infty$, and fix $T > 0$ so small that $F_* T^2/2 < R/4$. All mean trajectories considered below remain in $|q| < R/2$ for small $\hbar$.

There are constants $C_0, C_1 > 0$, depending on the fixed cutoff, such that for all x and all $q \neq 0$,

$$|F(x)| \le C_0\sqrt{|x|}, \qquad |F(x) - F(q)| \le C_1(1 + |q|^{-1/2})\,|x - q|. \tag{2}$$

For $|x - q| \le |q|/2$, the second bound follows from the derivative of the square-root force and the smooth cutoff. Otherwise use the first bound and $|x - q| \ge |q|/2$. This is a Lipschitz bound anchored at $q \neq 0$, **not** a global Lipschitz bound at the cusp.

For $0 < \hbar < e^{-1}$ set

$$a_\hbar = \frac{1}{\log(1/\hbar)}, \qquad g_{\hbar,k}(q) = (\pi\hbar)^{-1/4}\exp\left[-\frac{(q - a_\hbar)^2}{2\hbar} - \frac{ika_\hbar^{3/4}(q - a_\hbar)}{\hbar}\right]. \tag{3}$$

Choose a fixed $K > 0$ with

$$K^2 > \max\{16\sqrt{2}\,C_0,\ 16/3\}. \tag{4}$$

The two preparations are $g_{\hbar,0}$ and $g_{\hbar,K}$. Both have exactly the same Gaussian position density and $\operatorname{sd} Q = \operatorname{sd} P = \sqrt{\hbar/2}$. Their means are

$$(q, p) = (a_\hbar, 0), \qquad (q, p) = (a_\hbar, -Ka_\hbar^{3/4}), \tag{5}$$

both converging to $(0,0)$. The classical energies of these **centres** are respectively $-\frac{2}{3}a_\hbar^{3/2}$ and $(K^2/2-2/3)a_\hbar^{3/2}$: both vanish, but their signs differ.

Let $z_+(t)=(t^4/144,t^3/36)$ for $-T\le t\le T$, and let

$$z_\times(t)=\begin{cases}(t^4/144,t^3/36), & t\le 0,\\ (-t^4/144,-t^3/36), & t\ge 0.\end{cases} \tag{6}$$

The first trajectory returns to the positive side, while the second crosses to the negative side. Both follow the same incoming path, arrive at $(0,0)$ at $t=0$, and satisfy the cusp Newton equation.

Theorem. Let $\psi_{\hbar,k}(t)=e^{-itH_\hbar/\hbar}g_{\hbar,k}$. For $k=0,K$, the packet means and the position and momentum standard deviations converge uniformly on $[-T,T]$:

$$\begin{aligned}(\langle Q\rangle_{\hbar,0},\langle P\rangle_{\hbar,0}) &\longrightarrow z_+(t),\\ (\langle Q\rangle_{\hbar,K},\langle P\rangle_{\hbar,K}) &\longrightarrow z_\times(t),\\ \operatorname{sd}Q_{\hbar,k}(t)+\operatorname{sd}P_{\hbar,k}(t) &\longrightarrow 0.\end{aligned} \tag{7}$$

In particular, the Weyl expectations of every $b\in C_c^\infty(\mathbb{R}^2)$ converge uniformly to $b(z_+(t))$ and $b(z_\times(t))$, respectively.

2. A mean and spread estimate with an anchored force

Suppress the subscripts, let $q(t)=\langle Q\rangle$, $p(t)=\langle P\rangle$, and write $W(t)=\operatorname{sd}Q(t)+\operatorname{sd}P(t)$. The standard Ehrenfest identities and Cauchy–Schwarz give

$$\dot q=p,\qquad \dot p=\langle F(Q)\rangle,\qquad D^+W\le \operatorname{sd}P+\operatorname{sd}F(Q). \tag{8}$$

These identities hold as quadratic-form identities for the present $H_\hbar$: $V\in W^{1,\infty}$, the Gaussian lies in $D(Q)\cap H^2$, and the required weighted and H^1 domains are preserved by the bounded-force evolution. They can be obtained by regularizing Q,P and removing the cutoffs.

For $q(t)\neq 0$, (2) gives

$$\operatorname{sd}F(Q)\le\|[F(Q)-F(q)]\psi\|\le L(q)\operatorname{sd}Q,\quad L(q):=C_1(1+|q|^{-1/2}), \tag{9}$$

and also $|\langle F(Q)\rangle-F(q)|\le L(q)W$. Consequently

$$D^+W\le[1+L(q)]W,\qquad W(0)=\sqrt{2\hbar}. \tag{10}$$

On an interval reached from the initial time without a cusp crossing, if $|q|\ge a_\hbar/2$ and its length is at most $2T$, (10) gives

$$W(t)\le\sqrt{2\hbar}\,e^{C_T/\sqrt{a_\hbar}},\qquad \int L(q(t))W(t)\,dt\le C_T a_\hbar^{-1/2}\sqrt{\hbar}\,e^{C_T/\sqrt{a_\hbar}}=o(a_\hbar^N) \tag{11}$$

for every fixed $N>0$. Here and below C_T may change from line to line but is independent of $\hbar$. This super-polynomial smallness in $a_\hbar$ is why the logarithmic separation in (3) is convenient. After the controlled passage in §3, its additional factor $e^{Ca_\hbar^{-1/4}}$ can be absorbed into $e^{C_T/\sqrt{a_\hbar}}$, so the same form of bound applies on the outgoing interval.

On the positive side, $F(q)\ge 0$ within the chosen core. For $g_{\hbar,0}$ at positive times, or either preparation at negative times, (11) and (8) bootstrap $q(t)\ge a_\hbar/2$: the total force-bias error is $o(a_\hbar)$, whereas the force itself pushes away from the cusp in the appropriate time direction. On those intervals compare (q,p) with the classical solution from (5). The field is Lipschitz on $q\ge a_\hbar/2$ with constant $O(a_\hbar^{-1/2})$. Gronwall and (11) show that their phase-space distance is

$$O\Big(a_\hbar^{-1/2}\sqrt{\hbar}\,e^{C_T/\sqrt{a_\hbar}}\Big)=o(a_\hbar^N)\quad\text{for every fixed } N. \tag{12}$$

The displayed constant C_T absorbs the additional Gronwall factor.

3. The transmitted packet crosses the cusp

It remains to control the $k = K$ preparation while its mean passes through $q = 0$, where (11) cannot be used with a fixed cutoff. Start at $q = a_\hbar$ and $p = -Ka_\hbar^{3/4}$. Bootstrap, until the first arrival at $q = -a_\hbar$, the two inequalities

$$W \le a_\hbar, \qquad p \le -\frac{K}{2} a_\hbar^{3/4}. \tag{13}$$

While $|q| \le a_\hbar$, the first inequality and (2) imply

$$|\langle F(Q)\rangle| \le C_0 \sqrt{\mathbb{E}|Q|} \le C_0 \sqrt{2a_\hbar}. \tag{14}$$

The second inequality bounds the passage time by $4a_\hbar^{1/4}/K$. In that time the total change in p is at most $(4\sqrt{2}C_0/K)a_\hbar^{3/4}$, which is strictly below $(K/2)a_\hbar^{3/4}$ by (4).

The same velocity bound makes the singular coefficient integrable:

$$\int_{\text{passage}} |q(t)|^{-1/2}\, dt \le \frac{2}{Ka_\hbar^{3/4}} \int_{-a_\hbar}^{a_\hbar} |q|^{-1/2}\, dq = \frac{8}{K} a_\hbar^{-1/4}. \tag{15}$$

The mean crosses zero at most once, so (10) holds away from that single time; apply it on the two sides and use continuity at the crossing. It follows that

$$W \le \sqrt{2\hbar}\, e^{Ca_\hbar^{-1/4}} = o(a_\hbar). \tag{16}$$

The inequalities (13) therefore close by continuity. The exit time $\tau_\hbar$ satisfies $\tau_\hbar = O(a_\hbar^{1/4})$, and at exit

$$q(\tau_\hbar) = -a_\hbar, \qquad -\tfrac{3K}{2} a_\hbar^{3/4} \le p(\tau_\hbar) \le -\tfrac{K}{2} a_\hbar^{3/4}. \tag{17}$$

After this exit the force is negative in the chosen core. The same bootstrap as in §2 keeps $q \le -a_\hbar/2$ and makes (11)–(12) valid against the classical solution started at the **actual** exit mean (17). No self-adjoint Hamiltonian with a regularized potential is substituted for (1).

4. The classical limits and the mixtures

The zero-energy positive-side arrival solves $\dot q = -\sqrt{\frac{4}{3}q^{3/2}}$. Integration gives

$$|t| = \int_0^{q(t)} \frac{ds}{\sqrt{\frac{4}{3}s^{3/2}}} = 2\sqrt{3}\, q(t)^{1/4}, \qquad q(t) = \frac{t^4}{144}. \tag{18}$$

For the $k = 0$ centre, energy conservation gives on either side of $t = 0$

$$|t| = \int_{a_\hbar}^{q_a(t)} \frac{ds}{\sqrt{\frac{4}{3}(s^{3/2} - a_\hbar^{3/2})}}. \tag{19}$$

The part from $a_\hbar$ to $2a_\hbar$ is $O(a_\hbar^{1/4})$; on the remainder the integrand is bounded by $Cs^{-3/4}$ and tends to the integrand in (18). Thus (19) tends to (18).

For $k = K$ at negative times, and for its outgoing classical comparison after (17), the analogous integral has denominator

$$\sqrt{u_\hbar^2 + \frac{4}{3}(s^{3/2} - a_\hbar^{3/2})}, \qquad u_\hbar \asymp a_\hbar^{3/4}. \tag{20}$$

Condition (4) ensures $u_\hbar^2 \ge (4/3)a_\hbar^{3/2}$ in the outgoing case, so the integrand is dominated by $Cs^{-3/4}$; it tends to the one in (18). At the entrance for negative times, $u_\hbar$ is exactly $Ka_\hbar^{3/4}$

and the same argument applies. The exit-time shift tends to zero. Energy conservation gives the corresponding momentum limits, and the bounds in §§2–3 supply (7), including a uniform estimate on the shrinking passage interval.

The overlap of the two initial Gaussians is

$$\langle g_{\hbar,0}, g_{\hbar,K}\rangle = \exp\left[-\frac{K^2 a_\hbar^{3/2}}{4\hbar}\right] \longrightarrow 0. \tag{21}$$

It remains unchanged by unitary evolution. For $0 \le \lambda \le 1$ and fixed θ, normalize

$$\Psi_{\hbar,\lambda,\theta}(0) \propto \sqrt{\lambda}\, g_{\hbar,0} + e^{i\theta}\sqrt{1-\lambda}\, g_{\hbar,K}. \tag{22}$$

For any $b \in C_c^\infty(\mathbb{R}^2)$, the Fourier representation of Weyl quantization and $\|(e^{iA}-1)\psi\| \le \|A\psi\|$ show that

$$\|[\mathrm{Op}_\hbar^W(b) - b(z_k(t))]\psi_{\hbar,k}(t)\| \longrightarrow 0 \quad \text{uniformly for } |t| \le T, \tag{23}$$

using (7). Thus the cross term in a Weyl expectation equals $b(z_k(t))\langle\psi_{\hbar,0}, \psi_{\hbar,K}\rangle + o(1)$ and vanishes uniformly. The pure states (22) have the semiclassical limit

$$\lambda\delta_{z_+(t)} + (1-\lambda)\delta_{z_\times(t)}. \tag{24}$$

For $t \le 0$ this is a single point on the common incoming path; for $t > 0$ it has the prescribed two branch weights. The result does not assert that the finite-$\hbar$ Wigner function is nonnegative.

5. Consequence for STATE

The cusp's unique quantum evolution need not choose one classical continuation from the limiting incoming phase-space history. Even identical initial position densities admit distinct continuations through vanishing momentum offsets, and coherent superpositions can produce any mixture of the two demonstrated branches. These explicit preparations supplement the earlier point-merger obstruction and align with established rough-potential selection results; they do not derive a physical preparation rule, a waiting-time selection, or a necessary positive action constant.